

CMSC203 Assignment 2 Documentation

Class: CMSC203 CRN 31776

Program: Assignment #2

Instructor: Prof. Tanveer

Summary of Description: (A patient information GUI, learning Methods, Classes and Constructors)

Due Date: 02.20.2026

Integrity Pledge: I pledge that I have completed the programming assignment independently.

I have not copied the code from a student or any source.

Student's Name: ANDREA ONGOMEFEN

Part 1: Pseudo Code: Here is a pseudo code for Assignment 2 program:

CLASS Patient

FIELDS: firstName, middleName, lastName,
streetAddress, city, state, zipCode,
phoneNumber, emergencyName, emergencyPhone

CONSTRUCTORS:

Patient() → set all fields to ""
Patient(fName, mName, lName) → set name fields only
Patient(all 10 fields) → set everything

GETTERS: one for each field (returns field value)

SETTERS: one for each field (updates field value)

HELPERS:

fullName() → firstName + middleName + lastName
address() → street + city + state + zip
emergencyContact() → emergencyName + emergencyPhone

toString() →

"Patient Name: " + fullName()
"Address: " + address()
"Phone: " + phoneNumber
"Emergency Contact: " + emergencyContact()

END CLASS

CLASS Procedure

FIELDS: procedureName, procedureDate,
practitionerName, charges

CONSTRUCTORS:

Procedure() → set all fields to "" or 0.0
Procedure(name, date) → set name and date only
Procedure(all 4 fields) → set everything

GETTERS: one for each field (returns field value)

SETTERS: one for each field (updates field value)

toString() →

```

"Procedure: "    + procedureName
"Date: "        + procedureDate
"Practitioner: " + practitionerName
"Charges: $"    + charges

```

END CLASS

DISPLAY "The program was developed by Andrea Ongomefen 02/20/26"

Part 2: UML: Here is a (UML) for Assignment 2 program:



Part 3: Comprehensive Test Plan: Here is a test case for Assignment 2 program:

Cases	Input	Expected Output	Actual Output	Did Test Pass?
Case 1	" ", " " Test empty string patient	- doesn't crash - empty " " variables	- doesn't crash - empty " " variables	Y
Case 2	p.toString () test... pl.toStrin g()..	- Patient to string - Procedure to string	- Patient to string - Procedure to string	Y
Case 3	Total charges.. Testing getCharges	Patient to string Procedure toString (3x) Total charge output	Patient to string Procedure toString (3x) Total charge output	Y
Case 4	Test setters to " "	Patient values empty	Patient values empty	Y

Part 3: Screenshots related to the Test Plan:

Case 1

```
Welcome to the Patient Information System!

--This program demonstrates the use of the Patient and Procedure classes.--
--This program allows you to store and display patient information and procedure information.--

Patient Name:
Address: , ,
Phone:
Emergency Contact:
Emergency Phone:

Procedure Name: X-Ray
Procedure Date: 04/27/2026
Practitioner: Dr. Smith
Charges: 250.00

Procedure Name: Blood Test
Procedure Date: 04/27/2026
Practitioner: Dr. Jones
Charges: 150.00

Procedure Name: MRI
Procedure Date: 04/27/2026
Practitioner: Dr. Lee
Charges: 1200.00

Total Charges: $1,600.00
```

Case 2

```
<terminated> Demo [Java Application] /Users/andrea/p2/p4
Patient Name: John A. Doe
Address: 123 Main St, Anytown, CA 12345
Phone: 555-1234
Emergency Contact: Jane Doe
Emergency Phone: 555-5678
Procedure Name: Physical Exam
Procedure Date: 01/01/2024
Practitioner: Dr. Smith
Charges: $100
```

Case 3

```
Welcome to the Patient Information System!

--This program demonstrates the use of the Patient and Procedure classes.--
--This program allows you to store and display patient information and procedure information.--

Patient Name: Marcus J Williams
Address: 456 Elm Street, Houston, TX 77001
Phone: 713-555-2847
Emergency Contact: Diane Williams
Emergency Phone: 713-555-9931

Procedure Name: X-Ray
Procedure Date: 04/27/2026
Practitioner: Dr. Smith
Charges: 250.00

Procedure Name: Blood Test
Procedure Date: 04/27/2026
Practitioner: Dr. Jones
Charges: 150.00

Procedure Name: MRI
Procedure Date: 04/27/2026
Practitioner: Dr. Lee
Charges: 1200.00

Total Charges: $1,600.00
```

Case 4

```
<terminated> Demo [Java Application] /Users/andrea/p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_
Welcome to the Patient Information System!

--This program demonstrates the use of the Patient and Procedure classes.--
--This program allows you to store and display patient information and procedure information.--

Patient Name:
Address: , ,
Phone:
Emergency Contact:
Emergency Phone:
Total Charges: $1,600.00
```

Part 4: Lessons Learned:

I learned about setters and the importance of them while creating an object(I forgot and I gave me trouble). When creating constructors I realized the importance of "this." . How building objects can simplify data recording and the process behind making a class. This project was very interesting and enlightening because Java is a very straightforward language. It was simply to apply these ideas or methods like getters and setters.